Rule-Based Chatbot Report

Topic: College Helpdesk Chatbot

Language: Python 3 (if-elif-else rules)

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1. Introduction

A chatbot is a program that simulates conversation with users. This project implements a **rule-based chatbot** for a college helpdesk. It answers common student queries about admissions, courses, fees, library, hostel, exams, placements, timings and contact details. Unlike AI chatbots, it does not learn; it matches keywords in the user's message using **if-elif-else** conditions and returns a predefined answer.

2. Objectives

• Understand how rule-based conversational systems work.
• Apply Python string handling and conditional statements.
• Provide instant answers to frequently asked college questions.

3. Working / Algorithm

1. Display a welcome message.
2. Read user input in a loop.
3. Normalize input (convert to lowercase, remove punctuation).
4. Check keywords with if-elif conditions (e.g. 'fee', 'hostel', 'exam').
5. Print the matching response; if no rule matches, print a fallback message.
6. Exit when the user types 'bye', 'quit' or 'exit'.

4. Rules Implemented

Category	Trigger Keywords
Greeting	hello, hi, hey, namaste
Admission	admission, apply, enroll
Courses	course, branch, program, degree
Fees	fee, cost, payment
Library	library, book
Hostel	hostel, accommodation, room
Exams	exam, result, marks, grade
Timings	timing, time, open, hour
Contact	where, location, address, contact, phone
Faculty	faculty, teacher, professor
Placements	placement, job, package, company
Thanks / Bye	thank / bye, quit, exit

Fallback	anything else

5. Sample Chatbot Conversations

#	User Question	Chatbot Response
1	Hello	Hello! Welcome to GMIU College Helpdesk. How can I help you today?
2	How do I apply for admission?	Admissions open in June every year. You can apply online through the college website or visit the admission cell (Room 101, Admin Block). Required documents: 10th & 12th marksheets, ID proof, and 4 photos.
3	Which courses are available?	We offer B.Tech (CSE, IT, ECE, ME, CE), BBA, BCA, MBA and MCA. Which course would you like to know more about?
4	What is the fee for B.Tech?	The annual tuition fee is approximately Rs. 75,000 for B.Tech and Rs. 45,000 for BBA/BCA. Fees can be paid semester-wise online or at the accounts office. Scholarships are available for meritorious students.
5	What are the library timings for books?	The central library is open Monday–Saturday, 8 AM to 8 PM. Students can borrow up to 4 books for 15 days using their ID card. A digital library with e-journals is also available 24x7.
6	Is hostel facility available?	Separate hostels for boys and girls are available on campus. Hostel fee is Rs. 60,000 per year including mess. Rooms are allotted on a first-come, first-served basis. Contact the hostel warden office.
7	When are the exams and results?	Semester exams are held in December and May. Results are published on the student portal within 30 days. Revaluation forms must be submitted within 7 days of result declaration.
8	What are the college timings?	College timings are 9:00 AM to 4:30 PM, Monday to Friday. The campus is closed on Sundays and public holidays.
9	Where is the college located?	We are located at GMIU Campus, Sector 12, Gandhinagar, Gujarat. Phone: 079-1234-5678 | Email: info@gmiu.edu.in
10	Tell me about the faculty	We have 120+ experienced faculty members, most holding PhDs. You can view department-wise faculty profiles on the college website.
11	How are the placements?	Our placement cell records 85%+ placements. Top recruiters include TCS, Infosys, Wipro and Reliance. Highest package last year: 12 LPA, average package: 4.5 LPA.
12	Do you have a swimming pool?	Sorry, I didn't understand that. You can ask me about admissions, courses, fees, library, hostel, exams, placements, timings or contact details.
13	Thank you	You're welcome! Feel free to ask anything else about the college.
14	Bye	Goodbye! Best of luck with your studies. Visit us again!

6. Sample Terminal Output

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       GMIU COLLEGE HELPDESK CHATBOT
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Bot: Hello! I'm the College Helpdesk Bot.
Bot: Type 'bye' to exit.

You: Hi
Bot: Hello! Welcome to GMIU College Helpdesk. How can I help you today?
You: What is the hostel fee?
Bot: The annual tuition fee is approximately Rs. 75,000 for B.Tech and Rs. 45,000 for
     BBA/BCA. Fees can be paid semester-wise online or at the accounts office.
     Scholarships are available for meritorious students.
You: Placements?
Bot: Our placement cell records 85%+ placements. Top recruiters include TCS, Infosys,
     Wipro and Reliance. Highest package last year: 12 LPA, average package: 4.5 LPA.
You: Bye
Bot: Goodbye! Best of luck with your studies. Visit us again!
```

7. Advantages & Limitations

Advantages: simple, fast, predictable, no training data needed, easy to extend.
Limitations: understands only predefined keywords, cannot handle complex sentences, spelling mistakes or context from earlier messages.

8. Conclusion

The College Helpdesk Chatbot demonstrates how conditional logic in Python can build a useful conversational tool. It answers routine queries instantly and can be extended with more rules, a web interface, or NLP techniques in the future.